

Build an asynchronous AR reconciliation workflow engine using the provided dataset.

https://www.kaggle.com/datasets/asiryi/ai-powered-erp-ar-reconciliation?utm_source=chatgpt.com

The system must process records through modular stages: ingestion, matching, validation, and decision routing. Each stage may fail randomly and be retried. The workflow must resume from the last successful stage rather than restarting from scratch.

This is not an ML accuracy assignment.

The system should support parallel execution, persistence of workflow state, and safe handling of duplicate or repeated submissions.

Use Python and a backend-oriented stack of your choice. We care more about workflow orchestration, state management, retries, and system design than advanced matching logic.